Rank-27 Congruence ITE Sign-Mask Repair

Tactix Development Project

24 August 2026

Executive summary

This increment promotes a general representation repair for Congruence ITE extraction. At the exact conflict-9450 Rank-27 common state, current Tactix spent 630 ms in ITE extraction versus current instrumented Kissat's 117 ms. Tactix asked six exact ternary membership questions for each canonical small-clause record; Kissat obtained the same sign combinations from literal-local dense watches. The repair groups immutable records by their unsigned three-variable prefix and caches one byte containing the eight exact ternary sign patterns. Canonical iteration, candidate order, exact ticks and costs, scheduling, proof transactions, learning, pruning, and N1–N4 are unchanged.

The exact profile reduces the separately materialized membership lambda from 475.813M to 130.659M cycles, complete gate extraction from 983.981M to 714.124M, and whole-prefix cycles from 63.890B to 63.485B. All 64 event-trace rows are byte-identical. The complete frozen B28 plus fixed C5 guard passes 66/66 independently checked tasks, and exact-source ASan/UBSan native-LRAT fuzz passes 10,000 random cases with zero failures.

Checked Rank-27 job `1645956` verifies the byte-identical 1,514,395,111-byte DRAT proof and removes it. Every non-time statistic is exact; Congruence improves 16.910 to 16.154 seconds. Total internal time is 409.438 seconds versus 403.855 for the predecessor because unchanged-trajectory search/scaffolding is 5.259 seconds noisier. The intended owner improvement, exact common-state cycle reduction, and full correctness gates justify promotion. The 300-second Rank-27 threshold is an engineering milestone, not a stopping condition: convergence continues while portable current-Kissat gaps remain.

Frozen Kissat 4.0.4 and all other frozen reference solvers were not rerun. The user-canceled full Tier-D arrays remain frozen, non-scorable, and residue-free.

1 Authority and exact localization

The immutable input is `tierD24-26/5911653e4edadec391654c8f06353f9a-oski15a01b06s_opt.cnf.xz`, SHA-256 `5435b1b271b049ed9291ab1b9b297afb33b8d2d3c50ff1b48bfaed4ddf7bb4e9`. Every Rank-27 job uses two CPUs and the 16 GiB/1200 s contract. Frozen Kissat 4.0.4's accepted 278.187-second result is immutable and was not rerun. Current instrumented Kissat is source-observer evidence only, never scoring evidence.

Paired current/current observer `1645876` shares 9,450 conflicts and 5,795,248 decisions at the already frozen exact boundary. Tactix/current Kissat cumulative Congruence is 2,267/1,643 ms. A source-equivalent substage observer, `1645878`, reports 2,346/1,644 ms and separates gate types:

Substage	Tactix, ms	current Kissat, ms	Tactix/Kissat
AND extraction	283	568	0.498x
XOR extraction	89	269	0.331x
ITE extraction	630	117	5.385x
Complete Congruence	2346	1644	1.427x

Thus neither AND, XOR, general gate extraction, binary liveness, nor forward subsumption is the first owner. Canonical ITE collection performs six sign-flipped exact ternary probes per record. That repeated common structural work is the portable divergence.

An initial compact record-key hash did not solve it. After its mandatory crash repair, exact-state job `1645880` preserved the complete non-time image but worsened ITE 646 to 938 ms, Congruence 2355 to 2715 ms, and wall 16.93 to 17.57 seconds. Hashing and open-addressed random access displaced the cache-local first-literal range lookup. The experiment and its feature-specific test were rejected and reverted; its frozen incident and campaign manifests remain the authority.

2 Repair and mandatory failure workflow

After `ClauseIndex` construction, sorting, and deduplication, the accepted repair sorts a temporary vector of record ordinals by the unsigned first-three- variable prefix. Each group accumulates one bit for each exact size-three sign pattern and publishes that byte beside each record in original canonical order. ITE collection reads the byte only for distinct-variable records in that same prefix. Repeated-variable degeneracies and later cross-prefix four-clause candidate verification retain the authoritative first-literal range lookup. The temporary four-byte ordinal per record is released after construction; the scoped immutable index retains one byte per record.

The final source diff SHA-256 is `4c2cd4a60706dabe4b9fec7b7749750bce9b647e268d8e932680c2144034591a`. Release pin SHA-256 is `9e9f0dab6a5f3634100dd9a32b5262246a9d0f1d5590d0377c3142546436defa`; ASan/UBSan pin SHA-256 is `1e02dd4646d722a5236c18feb9d1f37d61bcc6b255a66ebb942016ea128aafb3`. No live-clause authority, proof-rule choice, strategy input, elapsed-time branch, benchmark identity, deadline, per-instance arm, or tuned constant is introduced.

Two unpromoted prototypes triggered the mandatory bounded Claude workflow. The compact hash first crashed when a work-budget stop left the index unfinished and its new directory empty. Claude restored the previous conservative false result and added a focused truncation regression; Codex independently reviewed the two-file diff and passed Release 8/8 plus focused ASan/UBSan 5/5. Because the hash was later negative, the feature and specific regression were removed. Incident `benchmarks/runs/20260824-1829-congruence-ternary-directory-unit-crash/` owns the frozen evidence.

The first sign-mask prototype retained the last record's mask during later candidate verification and rejected a valid ITE gate. Bounded Claude repair reset the collection-only state before verification. Independent review confirmed exact grouping, degenerate fallback, conservative truncated-index behavior, direct cross-prefix verification, and unchanged order/tick/schedule sites; Release 8/8 and focused ASan/UBSan 5/5 pass. Incident `benchmarks/runs/20260824-1840-congruence-sign-mask-ite-unit-failure/` owns that workflow. Neither failing prototype entered SLURM evidence or an accepted pin, and repository certificate/core residue is zero.

3 Exact common-state performance and parity

Stage job 1645882 preserves every non-time line. ITE falls 669 to 530 ms while index construction rises 131 to 181 ms, saving 89 ms across the intended combined phase; complete Congruence falls 2395 to 2336 ms. Whole elapsed time is noisy because search/scaffolding outside identical inprocessing is slower in that sample.

Evidence	Predecessor	Candidate	Ratio
ITE membership lambda, cycles	475,813,458	130,659,304	0.27460x
Complete gate extraction, cycles	983,980,740	714,124,083	0.72575x
Complete sampled cycles	63,890,309,956	63,484,607,915	0.99365x
Sampled Congruence, ms	2211	2122	0.95975x

Profile job 1645884 reports zero lost samples. The new ordinal sort costs 151,130,287 cycles, only 0.2381% of candidate total, and does not displace the membership reduction. Trace job 1645886 emits 64 untruncated rows, byte-identical to the frozen predecessor with SHA-256 7a6216bb86c236512dea03de76c4c9ab2ffdd745bd7ca7b9fa5507e708987cc1. This fixes rule and candidate order, exact tick/cost fields, scheduling order, learning, pruning, boundary state, and resulting state.

4 Correctness and regression gates

Release validation passes 73 of 74 tests. The only failure is the pre-existing opt-in documentation-size debt; every executable, behavioral, proof/model, cleanup, trace, API, N1, and rule/policy-boundary test passes. Focused Debug ASan/UBSan Congruence, random-policy, micro-proof, boundary, and reset tests pass with leak detection disabled because the execution wrapper prevents LeakSanitizer startup.

Full-width campaign benchmarks/runs/20260824-1852-BC-ite-sign-mask-guard/, array 1645888 and aggregate 1645954, validates both immutable arms on all 28 Tier-B and five inherited predeclared mechanism-relevant Tier-C inputs. Each arm passes 17 SAT models and 16 UNSAT proofs. There is no invalidity, solve loss, regression, cleanup failure, or certificate residue, and all 33 paired non-time statistic images are exact. Candidate PAR-2 is 1648.724 versus 1636.633 seconds, ratio 1.0073877; the small wall increase is outside Congruence with exact logical work. Summary SHA-256 is 4bc3326c21b8e0ff928aedda285a37322430c7ddd409450ef23295538f8a7488.

Exact-source fuzz job 1645887 completes in 01:11:06 with 10,000 random cases and zero failures: 2,628 checked SAT models, 3,490 checked native-LRAT UNSAT proofs, and 3,882 declared five-second nonterminations on which no oracle claim is made. Congruence fires in 3,570 verdict cases for 1,259,092 target applications. The 27-entry corpus has zero failures and its one predeclared nontermination is the Sweep-specific proof case. There is no sanitizer finding, rejected model/proof, crash, certificate residue, or job-temporary residue. Twenty-five retained timeout CNFs are declared diagnostic inputs, not certificates. Log SHA-256 is eeb8c0c741c5e15984343160146162741 3ce54684ba8cddc1b6eeef3eb1ab0c8.

5 Checked Rank-27 result and continuing disposition

Immutable campaign benchmarks/runs/20260824-1905-tierd-rank27-ite-sign-mask-witness/, job 1645956, completes on two CPUs in 00:08:45. Tactix reports UNSAT in 409.438 internal / 409.957 wall seconds with 643,976 KiB peak solver RSS. Pinned drat-trim reads the original CNF

and 1,514,395,111-byte proof, reports `s VERIFIED`, and exits zero after 112.415 internal / 112.588 wall seconds. Proof SHA-256 `0f99f55ac55571d6f83fe53ec8cb4a6a4c8df0a6e127442e50f47e25bea8bb2d` is byte-identical to predecessor `1645875`. Proof, CNF, checker derivatives, and scratch are deleted before atomic publication; cleanup flags are true and residue is zero.

Removing timing lines makes every solver statistic byte-identical, normalized SHA-256 `8cd3eda481799871a179e1483a57413622bc7832c09432df705b2308bafc2da7`. Congruence improves 16.910 to 16.154 seconds (0.955293x). Inprocessing differs by only +0.323 seconds, but unchanged-trajectory search/scaffolding is +5.259 seconds, producing the +5.583-second total. This broad node timing noise does not contradict the exact cycle and intended-owner evidence. The representation repair is accepted; the checked result remains above 300 seconds.

Work does not stop at the milestone. Residual exact common-state Congruence remains approximately 2.2–2.3 versus 1.64 seconds, including ITE near 530 versus 117 ms. Vivify is approximately 1.981/0.870 seconds and Substitute 1.489/0.780 at conflict 9450. BVE is substantially repaired but retains a broad complete-trajectory lead around 30–37/16.32 seconds after the common state has split. Search also retains a throughput lead and an unlocalized quality question. The next increment returns to the earliest source-backed common-state owner, preserving all Sweep, directed-bridge, BVE antecedent, proof-rule/strategy, reserved-trail, Congruence, and N1–N4 closures.